Comprehensive Technical & Educational Report: SPWM Generation and Safety

From Theoretical Foundations to Digital Implementation on STM32 Microcontrollers

Sinusoidal Pulse Width Modulation (SPWM) is the core of voltage and frequency control in industrial inverters. To deeply understand this system, we must traverse the path from signal theory to the physical limitations of digital processors (like STM32) and power modules (IPM). This report dissects this engineering journey in two distinct sections.

Part 1: Theoretical Foundations and SPWM Generation Mechanism

In the analog world and power electronics theory, the SPWM signal is generated by comparing two distinct waves. The diagram below illustrates this fundamental concept:

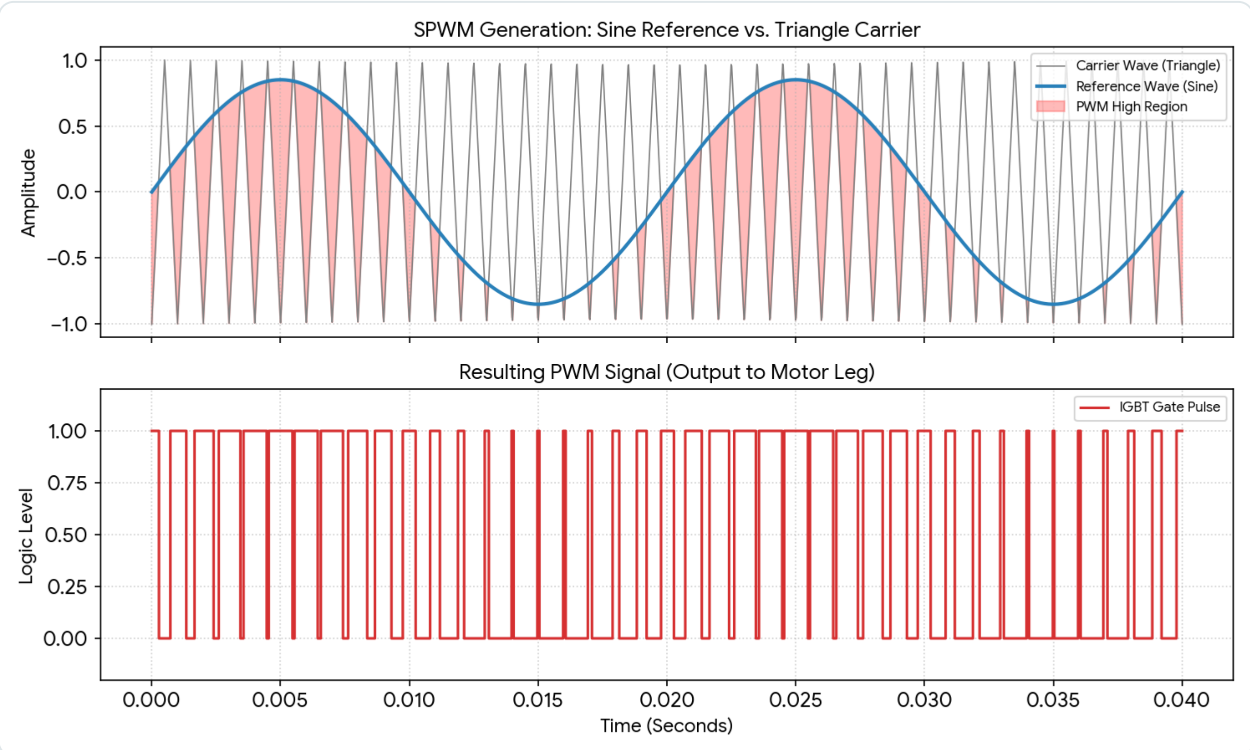

Figure 1: Comparison of the Reference Wave (Sine) and the Carrier Wave (Triangle) to generate PWM pulses.

As observed in the diagram above, the system consists of two main components:

- **Reference Wave:** A low-frequency sinusoidal signal (e.g., 50 Hz) representing the desired output voltage and frequency for the motor.
- **Carrier Wave:** A very high-frequency triangular wave (e.g., 20 kHz) responsible for determining the switching speed (turn-on and turn-off) of the power transistors.
- **Comparison Logic:** Whenever the amplitude of the sinusoidal wave is **greater** than the triangular wave, the output pulse transitions to a logical "High" state, turning on the IGBT gate. This continuous intersection generates a pulse train where the pulse width is maximum at the center of the half-cycle and minimum at the edges.

Part 2: Digital Implementation and Hardware Safety Limits

In the digital implementation on an STM32 microcontroller, the triangular wave is generated by the hardware timer counter (in Center-Aligned Mode), and the sinusoidal wave is modeled using a pre-calculated Look-Up Table (LUT) combined with the Direct Digital Synthesis (DDS) technique. However, in the real world, we cannot apply just any arbitrary pulse to the power module. The following image shows the digital simulation of these signals along with their protective boundaries:

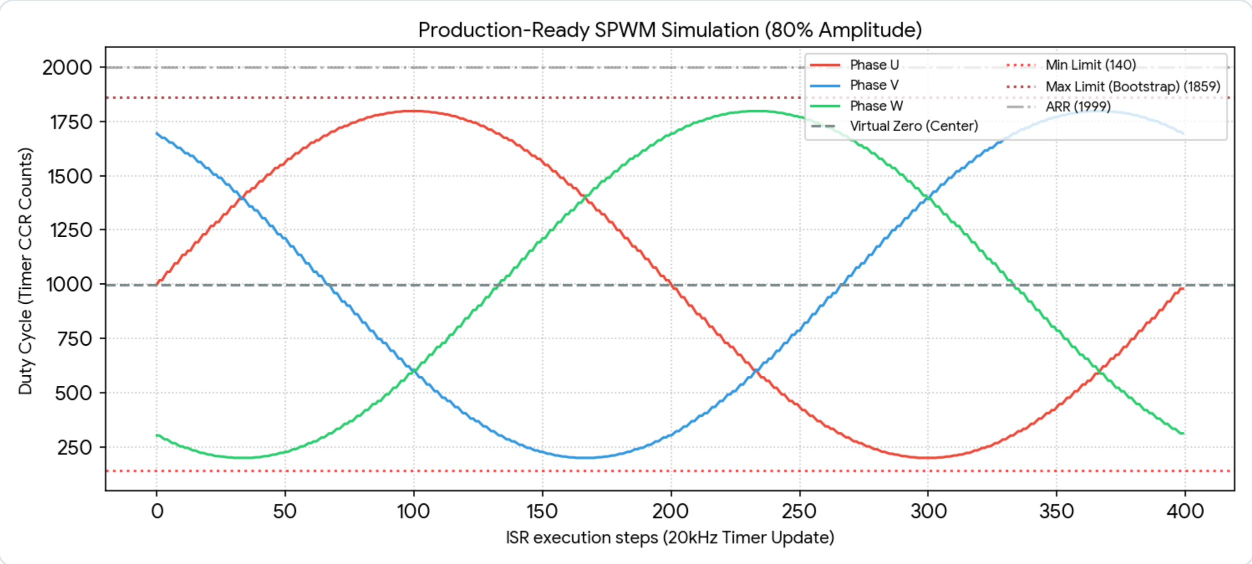

Figure 2: Digital Simulation of Three-Phase Duty Cycles with Hardware Limitations Applied (80% Amplitude).

In this simulation, executed at an **80%** amplitude, three perfectly symmetrical phases (U, V, W) with a 120-degree phase shift are generated. The vertical axis represents the Compare Register (`CCR`) values. Two critical boundaries are integrated into this architecture:

- **Lower Boundary (Minimum Pulse Limit = 140):** Indicated by the lower red dotted line. IGBT transistors require a minimum time to fully transition states. If the pulse generated by the microcontroller is too narrow (less than the sum of dead-time and gate delay), the IGBT enters the linear region and melts down. Our algorithm immediately clamps any pulse hitting this boundary to zero (0%).
- **Upper Boundary (Bootstrap Max Limit = 1859):** Indicated by the upper dark red dotted line. In bootstrap gate driver circuits, the high-side switch supply capacitor only charges when the low-side switch is turned on. Our protective algorithm prevents the duty cycle from reaching the timer ceiling (ARR = 1999) to guarantee the minimum time required to recharge the bootstrap capacitor.

> **Engineering Conclusion:**
> As is evident in the second image, at an 80% amplitude, the peak of the sine wave reaches exactly `1798`. Since this value sits safely below the critical bootstrap boundary (`1859`), the waveform is generated without any clipping, ensuring the highest harmonic quality and perfectly smooth motor operation.